

# HRMS LEAVE MANAGEMENT

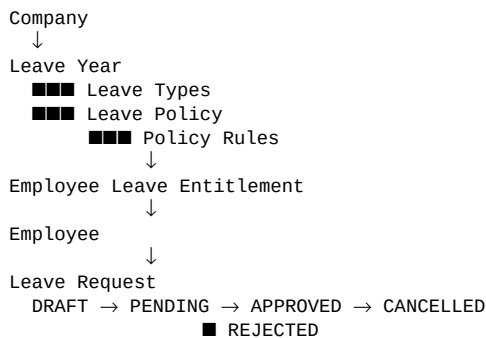
Complete Functional, Business & API Documentation

| Document | HRMS Leave Management Module                                                            |
|----------|-----------------------------------------------------------------------------------------|
| Version  | 1.0                                                                                     |
| Audience | HR • Admin • Developer • QA • Business Analyst                                          |
| API Base | <a href="http://localhost:5168/api">http://localhost:5168/api</a>                       |
| Scope    | Company → Leave Year → Leave Type → Policy → Entitlement → Request → Approval → Balance |

Purpose: A single reference for configuring leave, applying leave, approving/rejecting/cancelling requests, understanding balances, integrating APIs, and testing the workflow.

# 1. Executive Overview

The Leave Module manages the complete employee leave lifecycle. HR configures the organizational leave framework, assigns employee entitlements, employees create and submit requests, managers approve or reject them, and approved leave consumes balance. Cancellation of approved leave restores consumed balance.



## 2. Business Setup Order

| Step | Action               | Example                                |
|------|----------------------|----------------------------------------|
| 1    | Company              | ABC Technologies Pvt Ltd               |
| 2    | Leave Year           | FY2026: 01-Apr-2026 to 31-Mar-2027     |
| 3    | Leave Type           | CL - Casual Leave                      |
| 4    | Leave Policy         | Standard Employee Policy               |
| 5    | Policy Rule          | CL = 12 days                           |
| 6    | Employee Entitlement | EMP004 / CL / FY2026 = 12 days         |
| 7    | Leave Request        | EMP004 requests 2 days                 |
| 8    | Submit               | DRAFT → PENDING                        |
| 9    | Approve              | PENDING → APPROVED                     |
| 10   | Balance              | UsedDays increases by approved days    |
| 11   | Cancel if required   | APPROVED → CANCELLED; balance restored |

## 3. Core Entities and Relationships

| Entity                   | Purpose                                                               |
|--------------------------|-----------------------------------------------------------------------|
| Company                  | Organizational owner of HRMS data.                                    |
| LeaveYear                | Defines the period in which entitlement and leave requests are valid. |
| LeaveType                | Category such as CL, SL or EL.                                        |
| LeavePolicy              | Groups leave rules.                                                   |
| LeavePolicyRule          | Connects a policy/rule to a leave type and entitlement rule.          |
| EmployeeLeaveEntitlement | Employee-specific allocation for a leave year and leave type.         |
| LeaveRequest             | Employee leave application and workflow record.                       |
| LeaveRequestStatus       | DRAFT, PENDING, APPROVED, REJECTED, CANCELLED.                        |
| LeaveDayPart             | Defines full/partial-day boundaries.                                  |

## 4. Leave Year

A Leave Year defines the period during which an employee entitlement is valid. The request dates must be within the applicable leave year.

```
{
  "id": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8",
  "code": "FY2026",
  "name": "Financial Year 2026",
  "startDate": "2026-04-01",
  "endDate": "2027-03-31",
  "status": "ACTIVE"
}
```

- Leave Year must exist.
- Request dates must be valid for the leave year.
- Entitlement is unique for Employee + LeaveYear + LeaveType.

## 5. Leave Type

Leave Type defines the category of leave. Examples: CL = Casual Leave, SL = Sick Leave, EL = Earned Leave. A Leave Type alone does not create an employee balance.

```
{
  "id": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "code": "CL",
  "name": "Casual Leave",
  "isActive": true
}
```

## 6. Leave Policy and Policy Rule

A Leave Policy groups rules. A Policy Rule defines the rule applicable to a leave type.

```
{
  "id": "60000000-0000-0000-0000-000000000001",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "annualEntitlement": 12
}
```

The application validates that the selected Policy Rule belongs to the selected Leave Type.

## 7. Employee Leave Entitlement

EmployeeLeaveEntitlement represents the actual employee allocation.

```
{
  "id": "293BD6F3-934F-4147-99ED-08DEFD34A66C",
  "employeeId": "19ED1DF1-C3CC-4614-85F0-08DEF7B598A7",
  "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "entitledDays": 12.00,
  "usedDays": 0.00,
  "availableDays": 12.00
}
```

AvailableDays = EntitledDays - UsedDays

Note: the supplied domain entity contains EntitledDays. UsedDays/AvailableDays are shown here as the business balance projection used by the implemented transaction behavior; the physical database schema should match the deployed implementation.

## 8. Employee Entitlement APIs

| Method | Endpoint                           | Purpose   |
|--------|------------------------------------|-----------|
| POST   | /api/EmployeeLeaveEntitlement      | Create    |
| GET    | /api/EmployeeLeaveEntitlement      | List      |
| GET    | /api/EmployeeLeaveEntitlement/{id} | Get by ID |

| Method | Endpoint                           | Purpose |
|--------|------------------------------------|---------|
| PUT    | /api/EmployeeLeaveEntitlement/{id} | Update  |
| DELETE | /api/EmployeeLeaveEntitlement/{id} | Delete  |

## Create Entitlement

POST /api/EmployeeLeaveEntitlement

```
{
  "employeeId": "19ED1DF1-C3CC-4614-85F0-08DEF7B598A7",
  "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "leavePolicyRuleId": "9C56C99C-9192-43EE-351A-08DEFD343F02",
  "entitledDays": 12
}
```

201 Created  
"293BD6F3-934F-4147-99ED-08DEFD34A66C"

- Employee, Leave Year, Leave Type and Policy Rule must exist.
- Policy Rule must belong to the selected Leave Type.
- Duplicate Employee + LeaveYear + LeaveType is rejected.
- EntitledDays cannot be negative.

## 9. Leave Request Lifecycle

DRAFT

■■■ Update  
■■■ Delete  
■■■ Submit

↓

PENDING

■■■ Approve → APPROVED → Cancel → CANCELLED  
■■■ Reject → REJECTED

| Current   | Update | Delete | Submit | Approve | Reject | Cancel |
|-----------|--------|--------|--------|---------|--------|--------|
| DRAFT     | YES    | YES    | YES    | NO      | NO     | NO     |
| PENDING   | NO     | NO     | NO     | YES     | YES    | NO     |
| APPROVED  | NO     | NO     | NO     | NO      | NO     | YES    |
| REJECTED  | NO     | NO     | NO     | NO      | NO     | NO     |
| CANCELLED | NO     | NO     | NO     | NO      | NO     | NO     |

## 10. Leave Request Endpoint Summary

| Operation | Method | Endpoint                         |
|-----------|--------|----------------------------------|
| Create    | POST   | /api/leave-requests              |
| Get by ID | GET    | /api/leave-requests/{id}         |
| Get all   | GET    | /api/leave-requests              |
| Update    | PUT    | /api/leave-requests/{id}         |
| Delete    | DELETE | /api/leave-requests/{id}         |
| Submit    | POST   | /api/leave-requests/{id}/submit  |
| Approve   | POST   | /api/leave-requests/{id}/approve |
| Reject    | POST   | /api/leave-requests/{id}/reject  |
| Cancel    | POST   | /api/leave-requests/{id}/cancel  |

## 11. Create Leave Request

POST /api/leave-requests

```
{
  "employeeId": "19ED1DF1-C3CC-4614-85F0-08DEF7B598A7",
  "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "startDayPartId": "11111111-1111-1111-1111-111111111111",
  "endDayPartId": "22222222-2222-2222-2222-222222222222",
  "fromDate": "2026-08-25",
  "toDate": "2026-08-26",
  "reason": "Personal work"
}

201 Created
"410C17B0-632E-471D-6E42-08DEFDDFFA82"
```

The intended employee workflow is to create a DRAFT, review it, and then submit it.

## 12. Update Leave Request

PUT /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82

```
{
  "id": "410C17B0-632E-471D-6E42-08DEFDDFFA82",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "startDayPartId": "11111111-1111-1111-1111-111111111111",
  "endDayPartId": "22222222-2222-2222-2222-222222222222",
  "fromDate": "2026-08-25",
  "toDate": "2026-08-27",
  "reason": "Personal work - updated"
}

204 No Content
```

Only DRAFT requests can be updated. Route ID and command ID must match. Dates, day parts, leave year, overlap and TotalDays are validated/recalculated.

## 13. Delete Leave Request

DELETE /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82

204 No Content

Delete uses the shared soft-delete mechanism and is allowed only for DRAFT requests.

## 14. Submit Leave Request

POST /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82/submit

No request body.

204 No Content

Submit performs server-side revalidation and recalculates TotalDays, then changes DRAFT → PENDING. Overlap with another PENDING or APPROVED request is prevented.

## 15. Approve Leave Request

POST /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82/approve

```
{
  "id": "410C17B0-632E-471D-6E42-08DEFDDFFA82",
  "approvalReason": "Approved by manager"
}

204 No Content
```

Only PENDING requests can be approved. Approval checks available entitlement and then updates request status and balance transactionally.

BEGIN TRANSACTION

```

Validate PENDING
Find entitlement
Check AvailableDays >= TotalDays
UsedDays += TotalDays
Status = APPROVED
COMMIT

```

## 16. Insufficient Balance

```

Balance = 10
A = 10 days PENDING
B = 3 days PENDING
C = 1 day PENDING

```

```

Approve C:
UsedDays = 1
Available = 9

```

```

Try to approve A:
Requested = 10
Available = 9

```

```

10 > 9
=> 409 Conflict
=> A remains PENDING
=> UsedDays remains 1

```

Pending requests do not consume balance in the current workflow. Only approval consumes balance.

## 17. Reject Leave Request

```

POST /api/leave-requests/169DBDC0-3C77-4099-6E41-08DEFDDFFA82/reject

{
  "id": "169DBDC0-3C77-4099-6E41-08DEFDDFFA82",
  "rejectionReason": "Leave cannot be approved due to business requirements."
}

204 No Content

```

Only PENDING requests can be rejected. Rejection records RejectedOn and RejectionReason and does not consume leave balance.

## 18. Cancel Approved Leave

```

POST /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82/cancel

{
  "id": "410C17B0-632E-471D-6E42-08DEFDDFFA82",
  "cancellationReason": "Personal plans changed"
}

204 No Content

```

Only APPROVED requests can be cancelled. Cancellation restores the consumed balance in the same transactional workflow.

```

Before:
Entitled = 10
Used = 2
Available = 8

```

Cancel 2-day approved leave

```

After:
Entitled = 10
Used = 0
Available = 10

```

## 19. Get Leave Request

```

GET /api/leave-requests/410C17B0-632E-471D-6E42-08DEFDDFFA82

200 OK

```

```
{
  "id": "410C17B0-632E-471D-6E42-08DEFDDFFA82",
  "employeeId": "19ED1DF1-C3CC-4614-85F0-08DEF7B598A7",
  "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8",
  "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
  "status": "APPROVED",
  "fromDate": "2026-08-20",
  "toDate": "2026-08-20",
  "totalDays": 0.50,
  "reason": "Personal work",
  "appliedOn": "2026-08-20T10:30:00Z",
  "approvedOn": "2026-08-20T11:05:58Z",
  "approvalReason": "Approved by manager"
}
```

## 20. Get Leave Requests

GET /api/leave-requests?pageNumber=1&pageSize=20

200 OK

```
{
  "items": [
    {
      "id": "410C17B0-632E-471D-6E42-08DEFDDFFA82",
      "employeeId": "19ED1DF1-C3CC-4614-85F0-08DEF7B598A7",
      "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",
      "status": "APPROVED",
      "fromDate": "2026-08-20",
      "toDate": "2026-08-20",
      "totalDays": 0.50
    }
  ],
  "pageNumber": 1,
  "pageSize": 20,
  "totalCount": 1
}
```

## 21. Balance Rules

| Event   | Balance Effect        | Status    |
|---------|-----------------------|-----------|
| Create  | No consumption        | DRAFT     |
| Update  | No consumption        | DRAFT     |
| Submit  | No consumption        | PENDING   |
| Approve | UsedDays += TotalDays | APPROVED  |
| Reject  | No change             | REJECTED  |
| Cancel  | UsedDays -= TotalDays | CANCELLED |

AvailableDays = EntitledDays - UsedDays

## 22. HR User Manual

### Initial setup

- Create/verify Company.
- Create and activate Leave Year.
- Create Leave Types.
- Create Leave Policy.
- Create Policy Rules.
- Assign Employee Leave Entitlements.

### Employee applying

- Create a leave request.
- Select leave type, dates and day parts.
- Enter reason.
- Review calculated days.
- Submit the request.

## Manager/HR approval

- Open PENDING request.
- Review employee, type, dates and days.
- Approve or reject.
- Approval checks balance and consumes entitlement.
- Rejection leaves balance unchanged.

## Cancellation

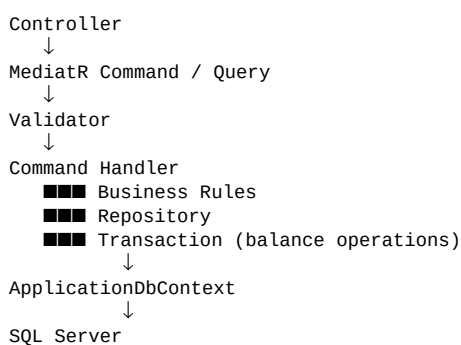
- Only APPROVED leave can be cancelled.
- Capture cancellation reason.
- Status becomes CANCELLED.
- Consumed balance is restored.

## 23. API Error Semantics

| HTTP | Meaning                                                |
|------|--------------------------------------------------------|
| 200  | Successful retrieval                                   |
| 201  | Created                                                |
| 204  | Successful state-changing operation with no body       |
| 400  | Invalid request / route mismatch / validation boundary |
| 404  | Entity/status/rule not found                           |
| 409  | Business rule or workflow conflict                     |

The exact global error JSON shape is controlled by the application's exception-handling middleware. Use the deployed middleware response as the definitive error contract.

## 24. Developer Architecture



Commands are separated into Command, Validator and Handler files. Business rules are centralized. Controllers orchestrate HTTP and do not contain leave business logic. Balance-changing operations use `ILeaveBalanceTransaction`.



## 25. Current Command Structure

```
Commands/
  ■■■ CreateLeaveRequest/
  ■ ■■■ CreateLeaveRequestCommand.cs
  ■ ■■■ CreateLeaveRequestCommandValidator.cs
  ■ ■■■ CreateLeaveRequestCommandHandler.cs
  ■■■ UpdateLeaveRequest/
  ■ ■■■ UpdateLeaveRequestCommand.cs
  ■ ■■■ UpdateLeaveRequestCommandValidator.cs
  ■ ■■■ UpdateLeaveRequestCommandHandler.cs
  ■■■ DeleteLeaveRequest/
  ■ ■■■ DeleteLeaveRequestCommand.cs
  ■ ■■■ DeleteLeaveRequestCommandValidator.cs
  ■ ■■■ DeleteLeaveRequestCommandHandler.cs
  ■■■ SubmitLeaveRequest/
  ■ ■■■ SubmitLeaveRequestCommand.cs
  ■ ■■■ SubmitLeaveRequestCommandValidator.cs
  ■ ■■■ SubmitLeaveRequestCommandHandler.cs
  ■■■ ApproveLeaveRequest/
  ■ ■■■ ApproveLeaveRequestCommand.cs
  ■ ■■■ ApproveLeaveRequestCommandValidator.cs
  ■ ■■■ ApproveLeaveRequestCommandHandler.cs
  ■■■ RejectLeaveRequest/
  ■ ■■■ RejectLeaveRequestCommand.cs
  ■ ■■■ RejectLeaveRequestCommandValidator.cs
  ■ ■■■ RejectLeaveRequestCommandHandler.cs
  ■■■ CancelLeaveRequest/
  ■ ■■■ CancelLeaveRequestCommand.cs
  ■ ■■■ CancelLeaveRequestCommandValidator.cs
  ■ ■■■ CancelLeaveRequestCommandHandler.cs
```

## 26. Data Integrity and Business Rules

- EmployeeLeaveEntitlement is unique for Employee + LeaveYear + LeaveType when not deleted.
- EntitledDays cannot be negative.
- Policy Rule must belong to the selected Leave Type.
- Leave request dates and day parts are revalidated before submission.
- TotalDays is calculated server-side.
- Overlapping PENDING/APPROVED leave is prevented.
- Approval consumes balance.
- Cancellation restores consumed balance.
- Balance-changing operations must be transactional.
- Approved, rejected and cancelled requests cannot be modified through the normal update workflow.

## 27. QA Test Matrix

| ID    | Scenario                          | Expected                                |
|-------|-----------------------------------|-----------------------------------------|
| LR-01 | Create valid request              | 201 Created                             |
| LR-02 | Update DRAFT                      | 204                                     |
| LR-03 | Submit DRAFT                      | DRAFT → PENDING                         |
| LR-04 | Approve with enough balance       | APPROVED + balance consumed             |
| LR-05 | Approve with insufficient balance | 409; remains PENDING; balance unchanged |
| LR-06 | Reject PENDING                    | REJECTED                                |
| LR-07 | Cancel APPROVED                   | CANCELLED + balance restored            |
| LR-08 | Update APPROVED                   | 409                                     |
| LR-09 | Delete PENDING                    | 409                                     |

| ID    | Scenario                 | Expected                  |
|-------|--------------------------|---------------------------|
| LR-10 | Approve REJECTED         | 409                       |
| LR-11 | Cancel REJECTED          | 409                       |
| LR-12 | Submit overlapping leave | 409                       |
| LR-13 | Duplicate entitlement    | 409                       |
| LR-14 | Negative entitlement     | Validation/business error |

## 28. Complete End-to-End Example

Employee: EMP004  
 Leave Year: FY2026  
 Leave Type: CL  
 Entitlement: 12 days

Initial:  
 Entitled = 12  
 Used = 0  
 Available = 12

1. Create 2-day request  
DRAFT
2. Submit  
PENDING
3. Approve  
APPROVED  
Used = 2  
Available = 10
4. Cancel approved request  
CANCELLED  
Used = 0  
Available = 12

## 29. Important Implementation Notes

- Route ID and command ID are checked for Update, Approve, Reject and Cancel.
- Submit requires only the request ID in the URL.
- Approval reason is optional in the current command contract but limited to 500 characters when supplied.
- Rejection and cancellation reasons are required and limited to 500 characters.
- Delete uses soft delete through AuditableEntity.
- Workflow timestamps use DateTimeOffset.UtcNow in the current handlers.
- The current workflow is single-level approval.
- Authentication/authorization is not part of this document's implemented scope unless enabled in the deployed API.

## 30. Quick Developer Reference

LEAVE REQUEST  
 POST /api/leave-requests  
 GET /api/leave-requests  
 GET /api/leave-requests/{id}  
 PUT /api/leave-requests/{id}  
 DELETE /api/leave-requests/{id}  
 POST /api/leave-requests/{id}/submit  
 POST /api/leave-requests/{id}/approve  
 POST /api/leave-requests/{id}/reject  
 POST /api/leave-requests/{id}/cancel

ENTITLEMENT  
 POST /api/EmployeeLeaveEntitlement  
 GET /api/EmployeeLeaveEntitlement

```
GET    /api/EmployeeLeaveEntitlement/{id}
PUT    /api/EmployeeLeaveEntitlement/{id}
DELETE /api/EmployeeLeaveEntitlement/{id}
```

## 31. Final Summary

The Leave Management module begins with Company and leave configuration, continues through Leave Year, Leave Type, Policy and Policy Rules, assigns employee entitlement, and then manages the employee request lifecycle. The critical balance rule is that PENDING requests do not consume entitlement, APPROVED requests consume entitlement, and CANCELLED approved requests restore entitlement. Invalid workflow transitions and insufficient balance are rejected.

This document is the baseline functional/API manual for the currently implemented module. Future features such as multi-level approval, accrual automation, carry-forward, encashment, holiday calendars, delegation, notifications and authorization should be documented as separate extensions when implemented.